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Studierichting: Informatica
Oefeningenreeks 1C nr. 48.8
$z^6 = -\sqrt{3} - i$

$$r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2$$

$$\tan\theta = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$$

we doen $+\ \pi$ omdat $-\sqrt{3} - i$ in het derde kwadrant zit

$$\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$$

$$-\sqrt{3} - i = 2\cos\frac{7\pi}{6} + 2i\sin\frac{7\pi}{6}$$

$$w_k = \sqrt[6]{2}\left(\cos\frac{\frac{7\pi}{6} + 2k\pi}{6} + i\sin\frac{\frac{7\pi}{6} + 2k\pi}{6}\right)$$

$$w_0 = \sqrt[6]{2}\left(\cos\frac{7\pi}{36} + i\sin\frac{7\pi}{36}\right)$$

$$w_1 = \sqrt[6]{2}\left(\cos\frac{19\pi}{36} + i\sin\frac{19\pi}{36}\right)$$

$$w_2 = \sqrt[6]{2}\left(\cos\frac{31\pi}{36} + i\sin\frac{31\pi}{36}\right)$$

$$w_3 = \sqrt[6]{2}\left(\cos\frac{43\pi}{36} + i\sin\frac{43\pi}{36}\right)$$

$$w_4 = \sqrt[6]{2}\left(\cos\frac{55\pi}{36} + i\sin\frac{55\pi}{36}\right)$$

$$w_5 = \sqrt[6]{2}\left(\cos\frac{67\pi}{36} + i\sin\frac{67\pi}{36}\right)$$

References